Residual conductivity from isotropic Dirac crossings

Abstract

We consider independent, spinless electrons in a two-dimensional periodic Schrödinger potential. Suppose that the Fermi energy meets the spectrum only at finitely many isolated, isotropic crossings of two bands. We prove that each crossing contributes 1/16 to the longitudinal conductivity, in atomic units, when a constant electric field is switched on, linear response is taken first, and the resulting current is averaged over long times. The proof separates three mechanisms: cancellation of the acceleration term by the filled bands, determination of the leading transition currents from the conical energies alone, and concentration of the averaged response near the crossings. In particular, the cone velocities cancel from the answer. Operator domains, local traces, and the passage to linear response are justified for real periodic H^1 potentials.

1 The result and its mechanism

A completely filled collection of bands cannot sustain a net acceleration current: translating the crystal momentum merely rearranges the occupied states. At a band crossing, however, an arbitrarily weak field can couple occupied and unoccupied states with arbitrarily small energy separation. The theorem below quantifies the current that survives this competition. It is conditional on the band geometry; no particular lattice symmetry is required once that geometry is assumed.

Let $\mathcal{R} \subset \mathbb{R}^2$ be a full-rank lattice, with a fundamental cell Ω, and let $\mathcal{B}$ be a fundamental cell of

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for all } R \in \mathcal{R}\}.$$

Thus $|\Omega|\,|\mathcal{B}| = (2\pi)^2$. We use atomic units and omit spin. For a real potential $V \in H^1_{\mathrm{per}}(\Omega)$, set

$$H = \tfrac{1}{2}p^2 + V, \qquad H_k = \tfrac{1}{2}(p+k)^2 + V, \qquad p = -\mathrm{i}\nabla. \tag{1.1}$$

The first operator acts on $L^2(\mathbb{R}^2)$; the fibers act on $L^2_{\mathrm{per}}(\Omega)$ with common domain $H^2_{\mathrm{per}}(\Omega)$. Write their eigenvalues, with multiplicity, as $\lambda_1(k) \le \lambda_2(k) \le \cdots$, and put $\lambda_0 = -\infty$.

Fix a Fermi energy μ_{F} and an integer $N \ge 1$. Our assumptions are that there are distinct points $k_1, \dots, k_M$ on the momentum torus $\mathbb{R}^2/\mathcal{R}^*$ such that

$$\lambda_N(k) < \mu_{\mathrm{F}} < \lambda_{N+1}(k) \quad \text{away from } k_1, \dots, k_M, \tag{1.2}$$

and, at every k_i,

$$\lambda_{N-1}(k_i) < \lambda_N(k_i) = \mu_{\mathrm{F}} = \lambda_{N+1}(k_i) < \lambda_{N+2}(k_i). \tag{1.3}$$

Each crossing is isotropic and untilted: for some $v_i > 0$,

$$\begin{aligned} \lambda_N(k_i + q) &= \mu_{\mathrm{F}} - v_i|q| + O(|q|^2), \\ \lambda_{N+1}(k_i + q) &= \mu_{\mathrm{F}} + v_i|q| + O(|q|^2). \end{aligned} \tag{1.4}$$

The remainders here are uniform in the direction of q.

Let $e_\beta, e_\alpha \in \mathbb{R}^2$ specify the field and current directions; they need not be unit vectors. A field of strength $\varepsilon \geq 0$ is switched on at time zero. The initial density matrix and its evolution are

$$\begin{aligned} \gamma_0 &= \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H), \qquad H_\varepsilon = H - \varepsilon x \cdot e_\beta, \\ \gamma_\varepsilon(t) &= e^{-\mathrm{i}tH_\varepsilon}\gamma_0 e^{\mathrm{i}tH_\varepsilon}, \qquad j_\varepsilon(t) = \underline{\mathrm{Tr}}\big((p \cdot e_\alpha)\gamma_\varepsilon(t)\big). \end{aligned} \tag{1.5}$$

Here H_ε is the self-adjoint closure of its restriction to $C_c^\infty(\mathbb{R}^2)$, and the trace per unit area of a periodic, locally trace-class operator is

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}(\mathbf{1}_\Omega A \mathbf{1}_\Omega). \tag{1.6}$$

Although the field Hamiltonian is not periodic, its density matrix is: a lattice translation changes H_ε only by a scalar.

Theorem 1. *Under* (1.2)–(1.4)*, the current in* (1.5) *is well defined. For every fixed* $t \geq 0$*, the linear response*

$$\sigma_{\alpha\beta}(t) := \lim_{\varepsilon \to 0^+} \frac{j_\varepsilon(t)}{\varepsilon}$$

exists, and

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T \sigma_{\alpha\beta}(t)\,\mathrm{d}t = \frac{M}{16}\, e_\alpha \cdot e_\beta. \tag{1.7}$$

The order of operations matters: this is a statement about the long-time average of the linear response, with the field strength sent to zero at each fixed time. It does not assert a long-time limit at a fixed nonzero field. For two crossings the spinless coefficient is $1/8$; two identical spin species would double it.

The proof follows the physical separation described above. In Section 2, we derive the response and show that the apparent term proportional to t cancels after integration over the filled bands. The remaining time average contains the nonnegative scalar kernel

$$\frac{1-\cos(T\Delta)}{T\Delta^3},$$

where Δ is an occupied-to-unoccupied energy gap. Gaps bounded away from zero contribute $O(T^{-1})$. Near a crossing, the relevant momenta have size T^{-1}. The only additional information needed there is the transition current, which Section 3 derives from the conical energies themselves. A rescaling then gives the universal constant in Section 4. The analytic foundations are collected in Appendix A so that they do not obscure these mechanisms.

2 From the switched field to interband transitions

Let

$$P(k) = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H_k), \qquad J_\eta(k) = (p+k)\cdot e_\eta \quad (\eta = \alpha, \beta).$$

Away from the crossings, $P(k)$ has rank N and is a smooth spectral projection. Its values at the finitely many crossings do not affect any direct integral. Throughout the proof, statements involving its rank or an occupied-to-unoccupied gap are understood away from those points.

Choose a measurable orthonormal eigenbasis $u_n(k)$ of H_k and set, for $n \leq N < m$,

$$\begin{aligned} \Delta_{mn}(k) &= \lambda_m(k) - \lambda_n(k), \\ Z_{nm}(k) &= \langle u_n(k), J_\alpha(k) u_m(k)\rangle \langle u_m(k), J_\beta(k) u_n(k)\rangle. \end{aligned} \tag{2.1}$$

Inner products are antilinear in the first argument. No differentiable choice of eigenvectors is needed. Every sum below is unchanged by a change of basis within a degenerate eigenspace.

We will repeatedly use one uniform bound. The occupied eigenvalues are bounded on the momentum torus, and the fiber graph norms are uniformly equivalent to the H^2 norm. Consequently

$$\sum_{n\le N} \|J_\eta(k)u_n(k)\|^2 \le C_\eta, \qquad \sum_{n\le N<m} |Z_{nm}(k)| \le C. \tag{2.2}$$

The second inequality is Cauchy–Schwarz and completeness of the eigenbasis. These bounds control the infinitely many unoccupied bands. The domain and graph-norm assertions used here are justified in Section A.1.

Proposition 2 (Interband response)**.** *The equilibrium current is zero. The fixed-time response is*

$$\sigma_{\alpha\beta}(t) = 2\int_{\mathcal{B}} \sum_{n\le N<m} \operatorname{Re} Z_{nm}(k)\, \frac{\sin(t\Delta_{mn}(k))}{\Delta_{mn}(k)^2} \frac{\mathrm{d}k}{(2\pi)^2}. \tag{2.3}$$

Its time average is therefore

$$\overline{\sigma}_{\alpha\beta}(T) := \frac{1}{T}\int_0^T \sigma_{\alpha\beta}(t)\,\mathrm{d}t = 2\int_{\mathcal{B}} \sum_{n\le N<m} \operatorname{Re} Z_{nm}(k)\, \frac{1-\cos(T\Delta_{mn}(k))}{T\Delta_{mn}(k)^3} \frac{\mathrm{d}k}{(2\pi)^2}. \tag{2.4}$$

Both integrals are absolutely convergent for finite t and $T > 0$.

2.1 An accelerating gauge

Multiplication by $G_\varepsilon(t) = e^{\mathrm{i}\varepsilon t x\cdot e_\beta}$ removes the linear potential. In this gauge the Bloch fibers evolve under $H_{k+\varepsilon t e_\beta}$. More precisely, let

$$\mathrm{i}\partial_t U_\varepsilon(k,t) = H_{k+\varepsilon t e_\beta} U_\varepsilon(k,t), \qquad U_\varepsilon(k,0) = I, \tag{2.5}$$

and write $R_\varepsilon(k,t) = U_\varepsilon(k,t)P(k)U_\varepsilon(k,t)^*$. The current becomes

$$j_\varepsilon(t) = \int_{\mathcal{B}} \operatorname{Tr}\left[(J_\alpha(k) + \varepsilon t\, e_\alpha\cdot e_\beta) R_\varepsilon(k,t)\right] \frac{\mathrm{d}k}{(2\pi)^2}. \tag{2.6}$$

The term proportional to εt is the explicit acceleration of the current observable. The change in R_ε will cancel its secular contribution.

Since

$$H_{k+\varepsilon s e_\beta} = H_k + \varepsilon s J_\beta(k) + \tfrac{1}{2}\varepsilon^2 s^2 |e_\beta|^2,$$

Duhamel's formula gives the first variation of the density matrix as

$$D(k,t) = -\mathrm{i}\int_0^t s\, e^{-\mathrm{i}(t-s)H_k}[J_\beta(k), P(k)] e^{\mathrm{i}(t-s)H_k}\,\mathrm{d}s. \tag{2.7}$$

For fixed time, differentiating (2.6) is legitimate uniformly in k; see Section A.3. Denote the resulting derivative at $\varepsilon = 0$ temporarily by $L_{\alpha\beta}(t)$. Evaluating (2.7) in the eigenbasis gives

$$L_{\alpha\beta}(t) = \int_{\mathcal{B}} \Bigg\{ tN\, e_\alpha\cdot e_\beta + 2\sum_{n\le N<m} \left[\operatorname{Re} Z_{nm}\left(\frac{\sin(t\Delta_{mn})}{\Delta_{mn}^2} - \frac{t}{\Delta_{mn}}\right) + \operatorname{Im} Z_{nm}\frac{1-\cos(t\Delta_{mn})}{\Delta_{mn}^2}\right]\Bigg\} \frac{\mathrm{d}k}{(2\pi)^2}. \tag{2.8}$$

For example, the sine term follows from $\int_0^t s\sin((t-s)\Delta)\,\mathrm{d}s = t/\Delta - \sin(t\Delta)/\Delta^2$. Only occupied-to-unoccupied matrix elements occur, because of the commutator with P.

2.2 Why the filled bands cancel the acceleration term

Consider the total occupied energy at a fixed momentum,

$$E(k) = \sum_{n=1}^{N} \lambda_n(k).$$

This is continuous and periodic everywhere, and equals $\mathrm{Tr}(H_k P(k))$ away from the crossings. There it is smooth even if occupied bands cross one another. Differentiating the spectral projection gives

$$\partial_\alpha E = \mathrm{Tr}(P J_\alpha), \qquad \partial_\beta \partial_\alpha E = N\, e_\alpha \cdot e_\beta - 2 \sum_{n \le N < m} \frac{\mathrm{Re}\, Z_{nm}}{\Delta_{mn}}, \tag{2.9}$$

where $\partial_\eta = e_\eta \cdot \nabla_k$. Indeed, $P(\partial_\beta P)P = (1-P)(\partial_\beta P)(1-P) = 0$, and differentiating $[H_k, P(k)] = 0$ yields

$$\langle u_m, \partial_\beta P\, u_n \rangle = -\frac{\langle u_m, J_\beta u_n \rangle}{\Delta_{mn}} \quad (n \le N < m).$$

This proves (2.9) without assumptions on degeneracies inside the occupied or unoccupied groups.

There is no boundary term hidden at a conical crossing. By (2.2), ∇E is bounded off the crossings. The smallest occupied-to-unoccupied gap is bounded below by $c|k - k_i|$ near k_i, so the right-hand side of (2.9) is locally integrable in two dimensions. Integrate on the momentum torus with disks of radius ρ removed. Each circular boundary has length $O(\rho)$ and carries a bounded gradient; its flux tends to zero. We obtain the filled-band sum rule

$$\int_{\mathcal{B}} N\, e_\alpha \cdot e_\beta \, \mathrm{d}k = 2 \int_{\mathcal{B}} \sum_{n \le N < m} \frac{\mathrm{Re}\, Z_{nm}(k)}{\Delta_{mn}(k)} \, \mathrm{d}k. \tag{2.10}$$

Thus the coefficient of t in (2.8) vanishes. This is precisely where the absence of a Fermi curve enters: isolated points leave no flux in this integration by parts.

The same argument applied to E itself gives $\int_{\mathcal{B}} \partial_\alpha E \, \mathrm{d}k = 0$ and hence $j_0 = 0$. Here E is bounded and continuous at the crossings, so its boundary integrals also vanish. Consequently $L_{\alpha\beta}(t)$ is the linear response $\sigma_{\alpha\beta}(t)$ in the theorem.

2.3 Time reversal removes the antisymmetric term

Complex conjugation $\mathcal{C}$ satisfies

$$\mathcal{C} H_k \mathcal{C} = H_{-k}, \qquad \mathcal{C} J_\eta(k) \mathcal{C} = -J_\eta(-k),$$

because V is real. Time-reversed eigenbases therefore have the same gaps and conjugate transition products:

$$\Delta_{mn}(-k) = \Delta_{mn}(k), \qquad Z_{nm}(-k) = \overline{Z_{nm}(k)}.$$

These identities may equivalently be read after summing over degenerate eigenspaces. The term containing $\mathrm{Im}\, Z_{nm}$ in (2.8) integrates to zero on the momentum torus. Together with (2.10), this proves (2.3).

For completeness, the estimates

$$\frac{|\sin(t\Delta)|}{\Delta^2} \le \frac{t}{\Delta}, \qquad \frac{1 - \cos(T\Delta)}{T\Delta^3} \le \frac{T}{2\Delta}$$

and (2.2) show absolute integrability: the only possible singularity is $|k - k_i|^{-1}$. They also justify the time integration in (2.4), completing the proof of Proposition 2.

3 What an isotropic cone says about current

The hypotheses describe energies, whereas the response also involves matrix elements of current. The bridge between them is the linear part of the Hamiltonian on the two-dimensional crossing subspace. Isotropy forces its two directional derivatives to satisfy fixed algebraic relations. These determine the transition strength without choosing phases for the two eigenvectors.

Proposition 3 (Transition current at a cone)**.** *Fix a crossing k_i, write $v = v_i$, and let $q = r\omega$ with $r > 0$ and $\omega \in S^1$. As $r \to 0$,*

$$\operatorname{Re} Z_{N,N+1}(k_i + r\omega) = v^2 \left[e_\alpha \cdot e_\beta - (e_\alpha \cdot \omega)(e_\beta \cdot \omega)\right] + O(r), \tag{3.1}$$

uniformly in ω.

Proof. Let $Q(q)$ be the spectral projection onto the two bands near μ_{F}. The separation from the remaining spectrum in (1.3) makes its contour-resolvent formula real analytic across $q = 0$, also as a map from L^2 to H^2; see [3, Section XII.2]. Projecting a fixed basis of $\operatorname{Ran} Q(0)$ and orthonormalizing gives an analytic frame $U(q) : \mathbb{C}^2 \to \operatorname{Ran} Q(q)$ with values in H^2_{per}. The reduced Hamiltonian $h(q) = U(q)^* H_{k_i+q} U(q)$ is an analytic Hermitian 2×2 matrix, with expansion

$$h(q) = \mu_{\mathrm{F}} I + A_q + O(|q|^2), \qquad A_\eta = \eta_1 A_1 + \eta_2 A_2 \quad (\eta \in \mathbb{R}^2). \tag{3.2}$$

Writing $U_0 = U(0)$ and differentiating at the crossing gives

$$A_\eta = U_0^* \big((p + k_i) \cdot \eta\big) U_0. \tag{3.3}$$

The derivatives of the frame cancel here because $H_{k_i} U_0 = \mu_{\mathrm{F}} U_0$ and $U(q)^* U(q) = I$.

For every η, the two eigenvalues of A_η are $\pm v|\eta|$, by (1.4) and (3.2). Consequently $\operatorname{tr} A_\eta = 0$ and $A_\eta^2 = v^2 |\eta|^2 I$. Polarization yields the decisive relation

$$A_\eta A_\zeta + A_\zeta A_\eta = 2v^2 (\eta \cdot \zeta) I \qquad (\eta, \zeta \in \mathbb{R}^2). \tag{3.4}$$

Thus the energy slopes already fix the symmetric products of the leading current matrices.

Let $p_\pm(q)$ be the two eigenprojections of $h(q)$. The gap is $2vr + O(r^2)$, so (3.2) gives

$$p_\pm(r\omega) = \pi_\pm(\omega) + O(r), \qquad \pi_\pm(\omega) = \tfrac{1}{2}(I \pm A_\omega / v), \tag{3.5}$$

uniformly in ω. For example, this follows directly by dividing $h(q) - \frac{1}{2} \operatorname{tr}(h(q)) I$ by half its eigenvalue gap. Also,

$$U(q)^* J_\eta (k_i + q) U(q) = A_{e_\eta} + O(r) \quad (\eta = \alpha, \beta).$$

The transition product in (2.1) is therefore

$$Z_{N,N+1}(k_i + r\omega) = \operatorname{tr}\big(\pi_-(\omega) A_{e_\alpha} \pi_+(\omega) A_{e_\beta}\big) + O(r).$$

To evaluate its real part, use that π_- has rank one and that (3.4) implies

$$\operatorname{Re} \operatorname{tr}(\pi_- A_{e_\alpha} A_{e_\beta}) = v^2 e_\alpha \cdot e_\beta, \qquad \operatorname{tr}(\pi_- A_{e_\eta}) = -v\, e_\eta \cdot \omega.$$

Substituting $\pi_+ = I - \pi_-$ now gives (3.1). □

In particular, for a single direction e, the leading transition strength is $v^2(|e|^2 - (e \cdot \omega)^2)$. It is the part of the current transverse to the radial momentum that couples the two bands. This geometric factor will become isotropic after angular integration.

4 Long-time averaging selects the crossings

We now prove Theorem 1. Choose disjoint disks of a fixed small radius ρ around the crossings on the momentum torus. Every occupied-to-unoccupied gap has a uniform positive lower bound, except for the pair $n = N$, $m = N+1$ inside these disks. This follows from compactness outside the disks and from (1.3) inside.

It follows from (2.2) and $|1 - \cos s| \le 2$ that all these transitions together contribute $O(T^{-1})$ to (2.4). Only the crossing pair in each disk remains. Fix one disk and abbreviate

$$\Delta(q) = \lambda_{N+1}(k_i + q) - \lambda_N(k_i + q), \qquad Z(q) = Z_{N,N+1}(k_i + q), \qquad v = v_i.$$

Its contribution is

$$I_i(T) = \frac{2}{(2\pi)^2} \int_{|q|<\rho} \operatorname{Re} Z(q) \frac{1 - \cos(T\Delta(q))}{T\Delta(q)^3} \, \mathrm{d}q. \tag{4.1}$$

The substitution $y = Tq$ identifies the surviving scale:

$$I_i(T) = \frac{2}{(2\pi)^2} \int_{|y|<T\rho} \operatorname{Re} Z(y/T) \frac{1 - \cos(T\Delta(y/T))}{(T\Delta(y/T))^3} \, \mathrm{d}y. \tag{4.2}$$

For every $y \neq 0$,

$$T\Delta(y/T) \longrightarrow 2v|y|, \qquad \operatorname{Re} Z(y/T) \longrightarrow v^2 \left[e_\alpha \cdot e_\beta - (e_\alpha \cdot \widehat{y})(e_\beta \cdot \widehat{y}) \right], \quad \widehat{y} = \frac{y}{|y|}.$$

These pointwise limits also determine the limit of the integral. Indeed, $\Delta(q)$ is bounded above and below by positive multiples of $|q|$, and $Z(q)$ is bounded. Extending the integrand by zero outside its disk, its absolute value is bounded by

$$C \min\{|y|^{-1}, |y|^{-3}\}, \tag{4.3}$$

using $1 - \cos s \le \min\{s^2/2, 2\}$. This bound is integrable on $\mathbb{R}^2$, both at the origin and at infinity. Dominated convergence, followed by polar coordinates $y = r\omega$, gives

$$\begin{aligned} \lim_{T\to\infty} I_i(T) = \frac{2}{(2\pi)^2} \int_{S^1} \left[e_\alpha \cdot e_\beta - (e_\alpha \cdot \omega)(e_\beta \cdot \omega) \right] \mathrm{d}\omega \\ \times \int_0^\infty \frac{v^2 r (1 - \cos(2vr))}{(2vr)^3} \, \mathrm{d}r. \end{aligned} \tag{4.4}$$

The angular integral is $\pi e_\alpha \cdot e_\beta$. In the radial integral, the change of variable $s = 2vr$ cancels the velocity entirely:

$$\int_0^\infty \frac{v^2 r (1 - \cos(2vr))}{(2vr)^3} \, \mathrm{d}r = \frac{1}{4} \int_0^\infty \frac{1 - \cos s}{s^2} \, \mathrm{d}s = \frac{\pi}{8}. \tag{4.5}$$

One direct evaluation of the last integral uses $s^{-2} = \int_0^\infty a e^{-as} \, \mathrm{d}a$: Tonelli's theorem gives

$$\int_0^\infty \frac{1 - \cos s}{s^2} \, \mathrm{d}s = \int_0^\infty a \left(\frac{1}{a} - \frac{a}{a^2 + 1} \right) \mathrm{d}a = \frac{\pi}{2}.$$

Combining (4.4) and (4.5),

$$\lim_{T\to\infty} I_i(T) = \frac{2}{(2\pi)^2} \, \pi e_\alpha \cdot e_\beta \, \frac{\pi}{8} = \frac{1}{16} \, e_\alpha \cdot e_\beta.$$

Summing over the M crossings, with the $O(T^{-1})$ remainder already estimated, proves (1.7). □

A Analytic foundations

This appendix justifies the operator statements used in the proof, including the fixed-time differentiation before any gap estimates are invoked. The latter point is useful: the closing gap does not obstruct linear response at a fixed time.

A.1 Domains and time evolution

In two dimensions a periodic H^1 potential is uniformly locally L^p for every finite p. Local Sobolev estimates and a partition into enlarged lattice cells give, for every $a > 0$,

$$\|Vf\|_2 \le a\|\Delta f\|_2 + C_a\|f\|_2, \qquad |\langle f, Vf\rangle| \le a\|\nabla f\|_2^2 + C_a\|f\|_2^2. \tag{A.1}$$

The first inequality is also the uniformly local perturbation bound of [3, Theorem XIII.96]. Both hold on $\mathbb{R}^2$ and on the periodic cell. By the Kato–Rellich theorem [2, Theorem X.12], H and H_k are self-adjoint on their H^2 domains. Their form domains are H^1, and their graph norms are equivalent to H^2, uniformly for k in a bounded set. The fiber resolvents are compact, and $k \mapsto H_k$ is analytic on the common domain. These observations also justify the smooth spectral projections used in the main text.

For the Stark operator, use the comparison operator

$$\mathcal{N} = H + |x|^2 + c,$$

with c sufficiently large. The harmonic-oscillator graph estimate and (A.1) show that $\mathcal{N}$ is self-adjoint on $H^2(\mathbb{R}^2) \cap \{f : |x|^2 f \in L^2\}$, with core $C_c^\infty(\mathbb{R}^2)$. Increasing c if necessary gives

$$\mathcal{N} \ge 1, \qquad \langle f, \mathcal{N} f\rangle \ge c_0(\|pf\|^2 + \|xf\|^2 + \|f\|^2).$$

Its graph norm controls $\|H_\varepsilon f\|$. On the core, the potential drops out of the commutator, and in the quadratic-form sense

$$\mathrm{i}[H_\varepsilon, \mathcal{N}] = p \cdot x + x \cdot p + \varepsilon p \cdot e_\beta.$$

Thus

$$\|H_\varepsilon f\| \le C_\varepsilon\|\mathcal{N} f\|, \qquad |\langle H_\varepsilon f, \mathcal{N} f\rangle - \langle \mathcal{N} f, H_\varepsilon f\rangle| \le C_\varepsilon\langle f, \mathcal{N} f\rangle.$$

The commutator theorem [2, Theorem X.37] proves essential self-adjointness of H_ε on C_c^∞.

The time-dependent operators $H_{k+\varepsilon t e_\beta}$ have the same domain H^2_{per}, and their dependence on time is continuously differentiable as maps from that domain to L^2. The common-domain propagator theorem [2, Theorem X.70] therefore supplies (2.5), with preservation of H^2. Its graph-norm bound is uniform on compact time intervals, for k in a fundamental cell and ε in a bounded interval. The same theorem applies to $\frac{1}{2}(p + \varepsilon t e_\beta)^2 + V$ on $L^2(\mathbb{R}^2)$; its propagator is the direct integral of the fiber propagators. Multiplying it by $G_\varepsilon(t)$ gives $e^{-\mathrm{i}tH_\varepsilon}$. This identity can be verified in the Schrödinger equation against C_c^∞ test functions and then extended by uniqueness for the self-adjoint generator. No domain assumption on $x\gamma_0$ is involved.

A.2 Local traces and normalization

Use the Bloch transform

$$(\mathcal{U} f)(k, x) = \sum_{R \in \mathcal{R}} e^{-\mathrm{i}k\cdot(x+R)} f(x + R), \qquad x \in \Omega,$$

with direct-integral measure $\mathrm{d}k/|\mathcal{B}|$ and the usual $L^2(\Omega)$ inner product in each fiber. It is unitary and transforms p into $p+k$ and H into H_k; see [3, Theorems XIII.97–XIII.99].

Here is the trace identity in the form needed above. For a periodic operator B with square-integrable Hilbert–Schmidt fiber norms, Parseval's identity in the lattice variable gives

$$\|\mathbf{1}_\Omega B\|_{\mathrm{HS}}^2 = \frac{1}{|\mathcal{B}|}\int_{\mathcal{B}} \|B(k)\|_{\mathrm{HS}}^2 \,\mathrm{d}k. \tag{A.2}$$

The corresponding right-localized identity follows by taking adjoints. If R is a periodic orthogonal projection with finite-rank fibers and finite fiber kinetic energy, both $R\mathbf{1}_\Omega$ and $\mathbf{1}_\Omega(p\cdot e_\alpha)R$ are Hilbert–Schmidt. Their product

$$\mathbf{1}_\Omega(p\cdot e_\alpha)R\mathbf{1}_\Omega = \big(\mathbf{1}_\Omega(p\cdot e_\alpha)R\big)\big(R\mathbf{1}_\Omega\big)$$

is trace class. The Hilbert–Schmidt pairing, or polarization of (A.2), yields

$$\frac{1}{|\Omega|}\operatorname{Tr}\big(\mathbf{1}_\Omega(p\cdot e_\alpha)R\mathbf{1}_\Omega\big) = \int_{\mathcal{B}} \operatorname{Tr}(J_\alpha(k)R(k))\frac{\mathrm{d}k}{(2\pi)^2}. \tag{A.3}$$

We use only the standard Hilbert–Schmidt and trace-class identities here; see [1, Section VI.6].

For $R_\varepsilon(t)$ in the accelerating gauge, the fiber ranks are N almost everywhere. The H^2 propagator bounds and the uniform H^2 bounds on occupied eigenvectors give finite fiber kinetic energy, uniformly on bounded time intervals. Moreover, $G_\varepsilon(t)$ commutes with $\mathbf{1}_\Omega$ and

$$G_\varepsilon(t)^*(p\cdot e_\alpha)G_\varepsilon(t) = p\cdot e_\alpha + \varepsilon t\, e_\alpha\cdot e_\beta.$$

Thus the same factorization proves local trace class for the physical current and gives exactly (2.6).

A.3 Differentiating the current at fixed time

Fix $t_0<\infty$ and $0\le\varepsilon\le 1$. All estimates in this subsection are uniform in $k\in\mathcal{B}$ and $0\le t\le t_0$. For $u\in H^2_{\mathrm{per}}$, write

$$\psi_\varepsilon(t) = U_\varepsilon(k,t)u, \qquad \psi_0(t) = e^{-\mathrm{i}tH_k}u.$$

The propagator bounds give $\sup_t\|\psi_\varepsilon(t)\|_{H^2}\le C_{t_0}\|u\|_{H^2}$. Duhamel's formula reads

$$\psi_\varepsilon(t)-\psi_0(t) = -\mathrm{i}\int_0^t e^{-\mathrm{i}(t-s)H_k}\left(\varepsilon s J_\beta(k) + \tfrac12\varepsilon^2 s^2|e_\beta|^2\right)\psi_\varepsilon(s)\,\mathrm{d}s. \tag{A.4}$$

It follows that the difference is $O(\varepsilon)$ in L^2. Interpolating with its uniform H^2 bound shows that it tends to zero in H^1. Dividing (A.4) by ε therefore gives convergence in L^2 to the first variation

$$\psi^{(1)}(t) = -\mathrm{i}\int_0^t s\, e^{-\mathrm{i}(t-s)H_k}J_\beta(k)e^{-\mathrm{i}sH_k}u\,\mathrm{d}s. \tag{A.5}$$

The convergence and its bounds are uniform for u in a bounded subset of H^2. In particular, they are uniform for all normalized occupied eigenvectors.

This is sufficient to differentiate a current expectation, despite the unbounded observable. Put $d_\varepsilon=\psi_\varepsilon-\psi_0$. Then

$$\begin{aligned}\frac{\langle\psi_\varepsilon, J_\alpha\psi_\varepsilon\rangle - \langle\psi_0,J_\alpha\psi_0\rangle}{\varepsilon} &= 2\,\mathrm{Re}\left\langle\frac{d_\varepsilon}{\varepsilon}, J_\alpha\psi_0\right\rangle \\ &\quad + \left\langle\frac{d_\varepsilon}{\varepsilon}, J_\alpha d_\varepsilon\right\rangle.\end{aligned} \tag{A.6}$$

The last term tends to zero because $d_\varepsilon/\varepsilon$ is bounded in L^2 and $d_\varepsilon \to 0$ in H^1. The first term converges by (A.5). Summing over the N occupied vectors and using the uniform bounds permits differentiation under the Bloch integral in (2.6).

Finally, the group $e^{-\mathrm{i}sH_k}$ preserves H^1, its form domain, uniformly for k in the cell. Since $J_\beta u \in H^1$ for $u \in H^2$, the first variations above also belong to H^1. Thus their current pairings agree with the trace expression (2.7). Expansion in an eigenbasis is justified by (2.2); the subsequent separation of terms containing inverse gaps is justified by the integrable conical bound used in Section 2. This proves all the fixed-time claims without imposing a uniform spectral gap.

References

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